

A MINI PROJECT REPORT



Submitted By:

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In partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

In

BIOTECHNOLOGY

DEPARTMENT OF BIOTECHNOLOGY

V.S.B ENGINEERING COLLEGE, KARUR.

(AN AUTONOMOUS INSTITUTION)

V.S.B ENGINEERING COLLEGE, KARUR.
(AN AUTONOMOUS INSTITUTION)

BONAFIDE CERTIFICATE

Certified that this mini project report entitled "**Network Pharmacology and Molecular Docking**" is the bonafide work of "**Poovitha.S (922525214023)**" who carried out the project work under my supervision during the academic year 2025 - 2026.

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Submitted for the Anna University Viva Voce held on

INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

We declare that the mini project report entitled " **Network Pharmacology and Molecular Docking** " submitted by me for the I Year Mini Project in Bachelor of Technology in Biotechnology is the original and independent work carried out in our college under the guidance of Dr. D Karthiga Ma'am, Assistant Professor, Department of Biotechnology, V.S.B. Engineering College, Karur. This document has not previously formed the basis for the award of any degree, diploma, fellowship, or similar academic title.

DATE:

PLACE: KARUR

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ACKNOWLEDGEMENT

We owe our sincere gratitude to V.S.B Engineering College for providing the facilities and supportive environment for the successful completion of our project titled “**Network Pharmacology and Molecular Docking**”

We express our heartfelt thanks to Dr V. S. Balsamy, Chairman of V.S.B Engineering College, for his valuable patronage and encouragement throughout our academic journey.

We are deeply grateful to Dr C. Vennila, Principal, and Dr T. S. Kiruba Shankar, Vice Principal, of V.S.B Engineering College, for their constant support and motivation in carrying out this project work.

We wish to express our profound gratitude to our Head of the Department, Dr M. Aniskumar, for providing us with the opportunity, guidance, and encouragement to undertake this project successfully.

We extend our special thanks to our project supervisor, Dr D Karthiga, whose inspiring guidance, valuable suggestions, continuous support, and encouragement helped us at every stage of this project. Her guidance greatly assisted us in preparing and completing this report successfully.

We also express our sincere thanks to all the faculty members of the Department of Biotechnology for sharing their valuable knowledge, technical support, and encouragement throughout the course of this work.

Our heartfelt appreciation goes to our friends and parents for their constant motivation, moral support, and help in every possible way during the completion of this project.

Finally, we express our deepest gratitude to everyone who contributed, directly or indirectly, to the successful completion of this project.

POOVITHA.S



V.S.B. ENGINEERING COLLEGE

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DEPARTMENT OF BIOTECHNOLOGY

Vision of the Institution

We endeavour to impart futuristic technical education of the highest quality to the student community and to inculcate discipline in them to face the world with self-confidence and thus we prepare them for life as responsible citizens to uphold human values and to be of service at large. We strive to bring of the Institution as an Institution of academic excellence of international standard.

Mission of the Institution

We transform persons into personalities by the state-of the art infrastructure, time consciousness, quick response and the best academic practices through assessment and advice.

Vision of the Department

To accomplish eminence in academic excellence through quality teaching with latest technology and innovations in the areas of agriculture, environment, health care and entrepreneurship for the welfare of the society.

Mission of the Department

❖ Disseminate a blending of knowledge acquisition and its application in real-life situations to the students.

- ❖ Equip the students to adapt to changing global and local needs through well designed curriculum and syllabus.
- ❖ Groom students to uphold professional ethics and develop leadership qualities
- ❖ Train students on issues related to social welfare

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

PEO 1 - Successful professional career and/or higher studies by gaining knowledge in fundamental mathematics and biological principles.

PEO 2 - Provide strong foundation in the core biotechnology courses to evaluate real life problems and to propose biotechnological solutions with economical and social viability.

PEO 3 - Sensitize on environmental, health and bioethical issues, Intellectual property rights, professional ethics and life-long learning through application orientated activities.

PROGRAM OUTCOMES(POs)

The Biotechnology Graduates will have the ability to

PO 1: Engineering Knowledge

Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem Analysis

Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences

PO3: Design/Development of Solutions

Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct Investigations of Complex Problems

Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Modern Tool Usage

Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6: The Engineer and Society

Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Environment and Sustainability

Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development. PO8: Ethics Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9: Individual and Team Work

Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication

Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project Management and Finance

Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12: Life-Long Learning Skill

Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOME

PSO I: Apply the knowledge in core and allied fields to solve complex engineering

PSO II : Able to expose their skills using latest tools to arrive cost effective and appropriate solutions.

PSO III : Apply the contextual knowledge with professional ethics to manage different projects in multi-disciplinary environment.

ABSTRACT

Ovarian cancer is one of the leading causes of cancer-related deaths among women worldwide due to its late diagnosis, high recurrence rate, and resistance to conventional chemotherapy. Natural products have gained significant attention as potential sources of novel therapeutic agents because of their diverse bioactive compounds and lower toxicity. *Anethum graveolens* (dill), a medicinal plant widely used in traditional medicine, contains several phytochemicals with antioxidant, anti-inflammatory, and anticancer properties. The present study employed an integrated network pharmacology and molecular docking approach to investigate the therapeutic potential of *Anethum graveolens* against ovarian cancer. Initially, phytochemical constituents of the plant were identified from public databases and screened for drug-likeness and pharmacokinetic properties using ADME analysis. Potential protein targets of the selected compounds and ovarian cancer-associated genes were collected from established databases. The overlapping targets were identified and used to construct a protein–protein interaction (PPI) network. Functional enrichment analyses, including Gene Ontology (GO) and Encyclopedia of Genes of disease and Genomes (KEGG) pathway analysis, were performed to understand the biological processes and signaling pathways involved. Finally, molecular docking studies were carried out to evaluate the binding affinity of the selected phytochemicals with key target proteins associated with ovarian cancer. The results demonstrated that the bioactive compounds of *Anethum graveolens* exhibited favorable pharmacokinetic properties, interacted with multiple therapeutic targets, and showed strong binding affinities toward key proteins involved in ovarian cancer progression. These findings suggest that *Anethum graveolens* possesses promising multitarget therapeutic potential against ovarian cancer. This study provides a scientific basis for further experimental validation.

and supports the development of plant-derived compounds as potential candidates for ovarian cancer treatment.

KEYWORDS:

Anethum graveolens, Nerol, Ovarian Cancer, Network Pharmacology, Molecular Docking, Phytochemicals, Bioactive Compounds, Protein–Protein Interaction (PPI), GeneCards, Cytoscape, KEGG Pathway Enrichment, AutoDock Vina, Drug Discovery, Target Prediction, Natural Products.

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LIST OF SYMBOLS AND ABBREVIATION:

- 1. ADME** - Absorption Distribution Metabolism and Excretion
- 2. BP**- Biological process
- 3. CC**- Cellular component
- 4. GO**- Gene Ontology
- 5. KEGG**- Kyto Encyclopedia of Genes and genomes
- 6. MF**- Molecular Function
- 7. PDB**- Protein Data Bank
- 8. SMILES**- Simplified Molecular Input Line Entry System
- 9. DL** – Drug-Likeness
- 10.PPI** – Protein-protein interaction
- 11.Kcal/mol** – Kilocalories per Mole

1. INTRODUCTION:

DISEASE:

Ovarian cancer is a disease in which abnormal cells grow uncontrollably in the ovaries and form malignant tumors. It is one of the most common and deadly gynecological cancers because it is often diagnosed at an advanced stage. Early symptoms are usually mild and may include abdominal bloating, pelvic pain, difficulty eating, frequent urination, and unexplained weight loss. Several factors contribute to the development of ovarian cancer, including genetic mutations, hormonal imbalance, age, family history, obesity, and environmental factors. The disease is associated with multiple molecular pathways that regulate cell growth, apoptosis, inflammation, and angiogenesis. Treatment options include surgery, chemotherapy, targeted therapy, and immunotherapy. However, drug resistance and adverse side effects remain major challenges, highlighting the need for safer and more effective therapeutic agents



Figure 1: image of ovarian cancer

PLANT:

Anethum graveolens L., commonly known as **dill**, is an aromatic medicinal herb belonging to the **Apiaceae** family. It is widely cultivated in Asia, Europe, and the Mediterranean region and has been used for centuries in traditional medicine as well as in culinary applications. The leaves and seeds of *Anethum graveolens* are rich in bioactive phytochemicals, including flavonoids, phenolic compounds, terpenoids, tannins, and essential oils. These compounds possess various pharmacological properties such as antioxidant, anti-inflammatory, antimicrobial, antidiabetic, hepatoprotective, and anticancer activities. Recent studies suggest that the phytochemicals present in *Anethum graveolens* may inhibit cancer cell growth by regulating cell proliferation, inducing apoptosis, reducing oxidative stress, and modulating multiple signaling pathways involved in cancer progression.



Figure 2: image of *Anethum graveolens*(dill)

NETWORK PHARMACOLOGY:

Network pharmacology is a modern computational approach used to study the interactions between bioactive compounds, target genes, proteins, and

disease-related pathways. Unlike the traditional "one drug–one target" approach, network pharmacology follows a "multi-component–multi-target–multi-pathway" strategy, making it especially suitable for studying medicinal plants that contain numerous active phytochemicals. In this study, network pharmacology was employed to investigate the therapeutic potential of *Anethum graveolens* against ovarian cancer. The phytochemicals present in the plant were collected from public databases and screened based on their pharmacokinetic properties. Potential target genes for the selected compounds and ovarian cancer-associated genes were then identified. The common target genes were analyzed using protein–protein interaction (PPI) networks, Gene Ontology (GO) enrichment, and KEGG pathway analysis to determine the biological processes and signaling pathways involved in the disease.

2. AIM:

To investigate the therapeutic potential of *Anethum graveolens* against ovarian cancer using network pharmacology and molecular docking by identifying its bioactive phytochemicals, predicting their target genes, analyzing the associated biological pathways, and evaluating the binding interactions between the selected compounds and key ovarian cancer-related proteins.

3. PRINCIPLE:

- Network Pharmacology

Network pharmacology is an integrated approach that studies the interactions among drugs, genes, proteins, and diseases. It helps identify multiple targets and pathways involved in disease treatment and provides a systematic understanding of drug action.

- Molecular Docking

Molecular docking is a computational method used to predict the binding interaction between a ligand and a target protein. The docking score indicates the binding affinity and stability of the protein-ligand complex.

4. OBJECTIVES:

1. To identify bioactive compounds present in dill plant
2. To collect Ovarian cancer-related target genes.
3. To identify common targets between dill compounds and ovarian cancer
4. To construct a protein-protein interaction network.
5. To perform Gene Ontology (GO) and KEGG pathway enrichment analyses.
6. To validate the interaction between dill compounds and ovarian cancer proteins through molecular docking.

5. TOOLS AND SOFTWARE USED:

1. Software/Database| Purpose
2. PubChem| Retrieval of dill compound structures
3. GeneCards| Collection of ovarian cancer genes
4. STRING| Protein-protein interaction analysis
5. Cytoscape| Network visualization
6. KEGG| Pathway enrichment analysis
7. Protein Data Bank (PDB)| Protein structure retrieval
8. AutoDock | Molecular docking

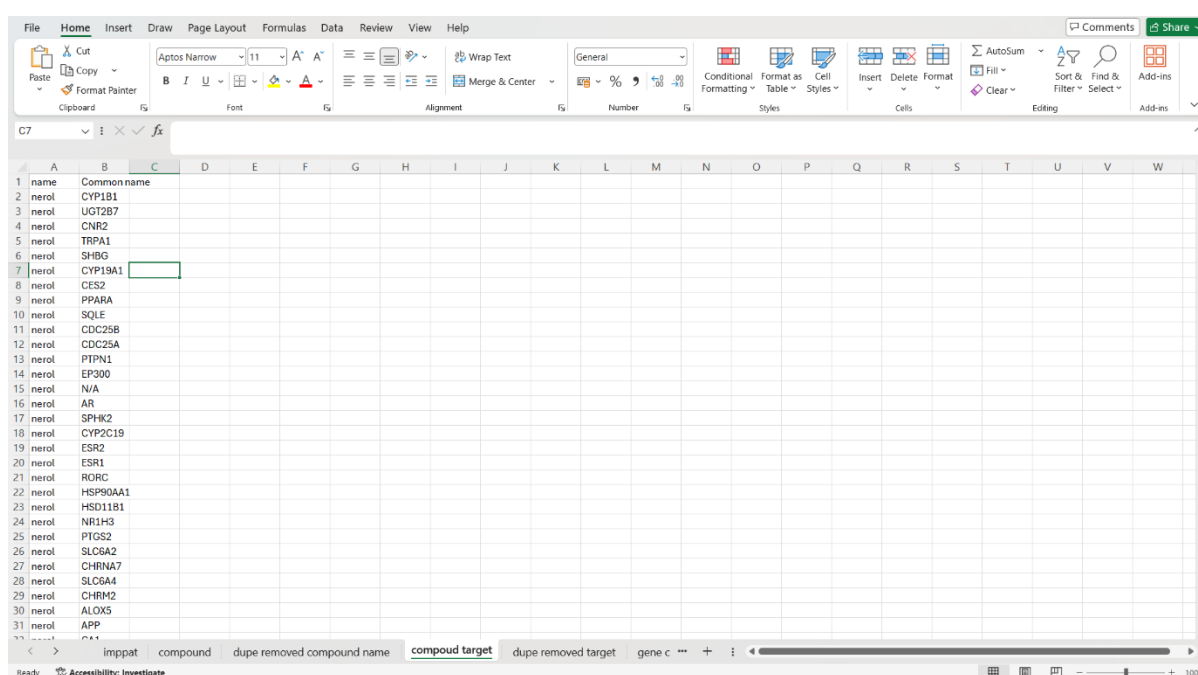
6. PROCEDURE:

Step 1: Identification of phytochemical compounds in *Anethum graveolense*

Major bioactive compounds such as 6-Gingerol, 6-Shogaol, and Zingerone were identified from literature and public databases. The phytochemical

compounds present in *Anethum graveolens* (dill) were identified using phytochemical databases and published scientific literature. These databases provide detailed information on the chemical constituents of medicinal plants, including their molecular structures and physicochemical properties.

Collect SMILES from pubchem web by entering the compounds name, collect SMILES and pubchem ID for all the compounds.



name	Common name
nerol	CYP1B1
nerol	UGT2B7
nerol	CNR2
nerol	TRPA1
nerol	SHBG
nerol	CYP19A1
nerol	CES2
nerol	PPARA
nerol	SQLE
nerol	CDC25B
nerol	CDC25A
nerol	PTPN1
nerol	EP300
nerol	N/A
nerol	AR
nerol	SPHK2
nerol	CYP2C19
nerol	ESR2
nerol	ESR1
nerol	RORC
nerol	HSP90AA1
nerol	HSD11B1
nerol	NR1H3
nerol	PTGS2
nerol	SLC6A2
nerol	CHRNA7
nerol	SLC6A4
nerol	CHR2
nerol	ALOX5
nerol	APP

Figure 3: Screenshot of table of phytochemical list

Step 2: Collection of Compound Targets

Potential targets of dill bioactive compounds were obtained from online target prediction databases. Here compound targets of nerol is collected which is found by ADME screening. Nerol is selected after checking some factors like molecular weight, hydrogen donor and acceptor and etc.. The

name	Common name
nerol	CYP1B1
nerol	UGT2B7
nerol	CYP2C8
nerol	TRPA1
nerol	SHBG
nerol	CYP19A1
nerol	CES2
nerol	PPARA
nerol	SQLE
nerol	CDC25B
nerol	CDC25A
nerol	PTPN1
nerol	EP300
nerol	N/A
nerol	AR
nerol	SPH2
nerol	CYP2C19
nerol	ESR2
nerol	ESR1
nerol	RORC
nerol	HSP90AA1
nerol	HSD11B1
nerol	NR1H3
nerol	PTGS2
nerol	SLC6A2
nerol	CHRNA7
nerol	SLC6A4
nerol	CHRM2
nerol	ALOX5
nerol	APP

Figure 4: Screenshot showing predicted compound targets

Step 3: Collection of Ovarian cancer Targets

Ovarian cancer -associated genes were collected from GeneCards database. GeneCards is a comprehensive human gene database that provides information on genes associated with diseases, their functions, and relevance scores. The identified disease-related targets were used for network pharmacology analysis.

PROCEDURE:

1. The keyword "Ovarian Cancer" was entered into the GeneCards database.
2. The list of genes associated with ovarian cancer was retrieved.
3. Genes with significant relevance to ovarian cancer were selected based on their relevance scores.
4. Duplicate and irrelevant entries were removed.
5. The final list of ovarian cancer-related target genes was saved for overlap analysis with the predicted targets of nerol.

Symbol	Name	Type	Relevance	Knowledge
BRCA1	BRCA1 DN Protein Co		552.6618	100
BRCA2	BRCA2 DN Protein Co		502.9339	100
PTEN	Phosphatase Protein Co		389.4234	100
TP53	Tumor Pro Protein Co		388.4562	100
MUC16	Mucin 16, f Protein Co		373.799	90
MLH1	MutL Hom Protein Co		360.9039	100
CHEK2	Checkpoint Protein Co		358.4112	100
ERBB2	Erb-B2 Rec Protein Co		355.9824	100
KRAS	KRAS Prot Protein Co		340.9429	100
CDH1	Cadherin 1 Protein Co		338.8178	100
CTNNB1	Catenin B Protein Co		335.9894	100
PIK3CA	Phosphatase Protein Co		335.4753	100
AKT1	AKT Serine Protein Co		332.6718	100
CDKN2A	Cyclin Dep Protein Co		329.1388	100
EGFR	Epidermal Protein Co		328.1503	100
ESR1	Estrogen R Protein Co		326.8258	100
MSH2	MutS Hom Protein Co		320.1438	100
ATM	ATM Serine Protein Co		310.9531	100
NBN	Nibrin Protein Co		304.8179	100
STK11	Serine/Thr Protein Co		299.9926	100
FOXO1	Forkhead f Protein Co		299.365	91
SMAD4	SMAD Fam Protein Co		294.1519	100
AR	Androgen f Protein Co		292.6332	100
BARD1	BRCA1 Ass Protein Co		289.4575	100
MSH6	MutS Hom Protein Co		289.3862	100
PALB2	Partner An Protein Co		281.0889	97
RAD51	RAD51 Rec Protein Co		275.1696	100
BRIP1	BRCA1 Inte Protein Co		275.0741	100
TERT	Telomerase Protein Co		275.0092	100
KIT	KIT Proto-C Protein Co		274.1791	100
MDM2	MDM2 Prot Protein Co		273.6638	100

Figure 5: Screenshot of GeneCards search results for Ovarian Cancer

Step 4: Identification of Common Targets

Open the website Venny 2.0 where the targets of *Anethum graveolense* compounds were compared with Ovarian cancer-related genes to identify overlapping targets. A Venn diagram is generated with which we can identify the common targets for the disease and the plant chosen.

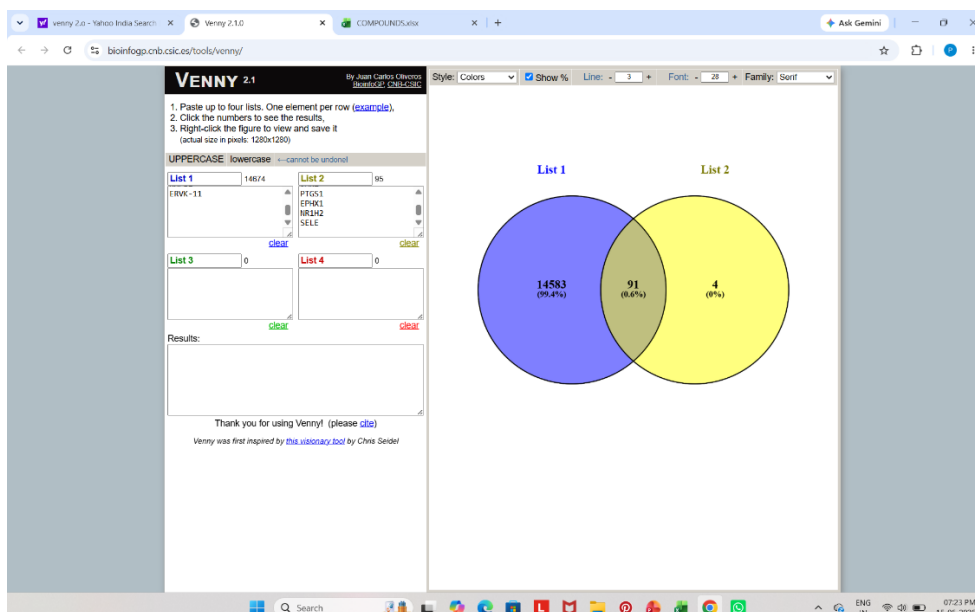


Figure 6: Venn diagram showing common targets

Step 5: Protein-Protein Interaction (PPI) Network Construction

The common targets were uploaded to the STRING database to generate a protein-protein interaction network. In the web page the organism is set to Homo sapiens and the protein interaction network is developed.

PROCEDURE:

- 1.The common target genes obtained from the overlap analysis were collected.
- 2.These genes were uploaded to the STRING database for PPI analysis.
- 3.The organism was set to Homo sapiens, and the default confidence score was applied.
- 4.The interaction network was generated and downloaded.
- 5.The PPI network was imported into Cytoscape for visualization and further analysis.
- 6.Hub proteins were identified based on network topology using Cytoscape tools.

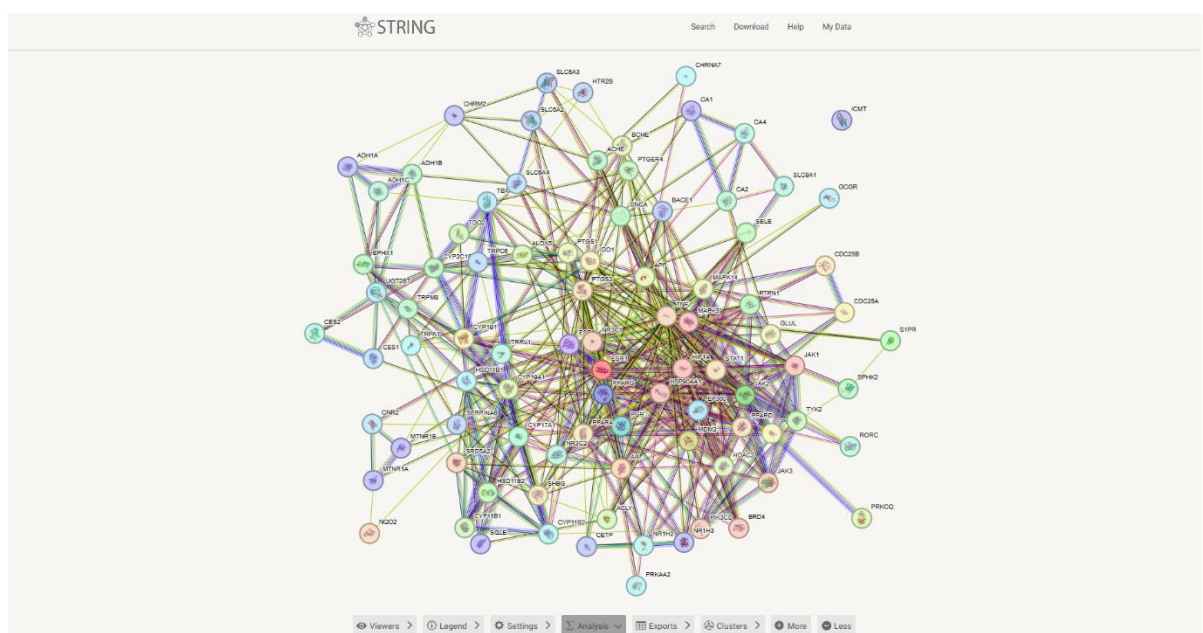


Figure 7: protein-protein interaction network

Step 6: Network Visualization Using Cytoscape

The PPI network was imported into Cytoscape software to identify hub genes and key target proteins. Cytoscape is an open-source software platform used to visualize and analyze complex biological networks. In this study, Cytoscape was used to visualize the interaction network of the common target genes associated with nerol and ovarian cancer, enabling the identification of key hub genes and their relationships within the network.

PROCEDURE:

1. The list of common target genes was obtained from the overlap analysis.
2. The interaction network generated from the STRING database was downloaded in a compatible format.
3. The network file was imported into Cytoscape software.
4. The network was visualized, and node and edge properties were adjusted for better representation.
5. Network topology analysis was performed to identify hub genes based on parameters such as degree, betweenness, and closeness centrality.
6. The finalized network image was exported for further interpretation and inclusion in the report.

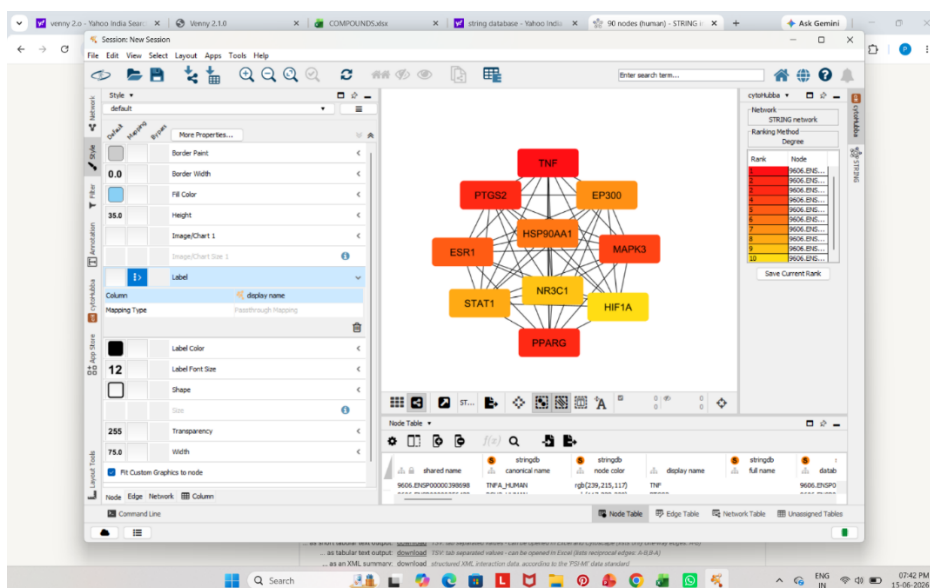


Figure 8: Cytoscape network analysis

Step 7: KEGG Pathway Enrichment Analysis

KEGG (Kyoto Encyclopedia of Genes and Genomes) pathway enrichment analysis is used to identify the biological pathways significantly associated with the common target genes. This analysis helps reveal the molecular mechanisms and signaling pathways through which nerol may exert its therapeutic effects against ovarian cancer. Websites like shinyGO can be used for generating KEGG pathway by choosing Homo sapiens and entering the common targets.

A lollipop plot is a graphical representation used to display the significance and enrichment of biological pathways or genes. In network pharmacology studies, it is commonly used to visualize the top enriched KEGG pathways or Gene Ontology (GO) terms, making it easier to compare their enrichment scores and statistical significance. Lollipop plot can be found in string database for

- KEGG pathway
- Biological process
- Molecular function
- Cellular components

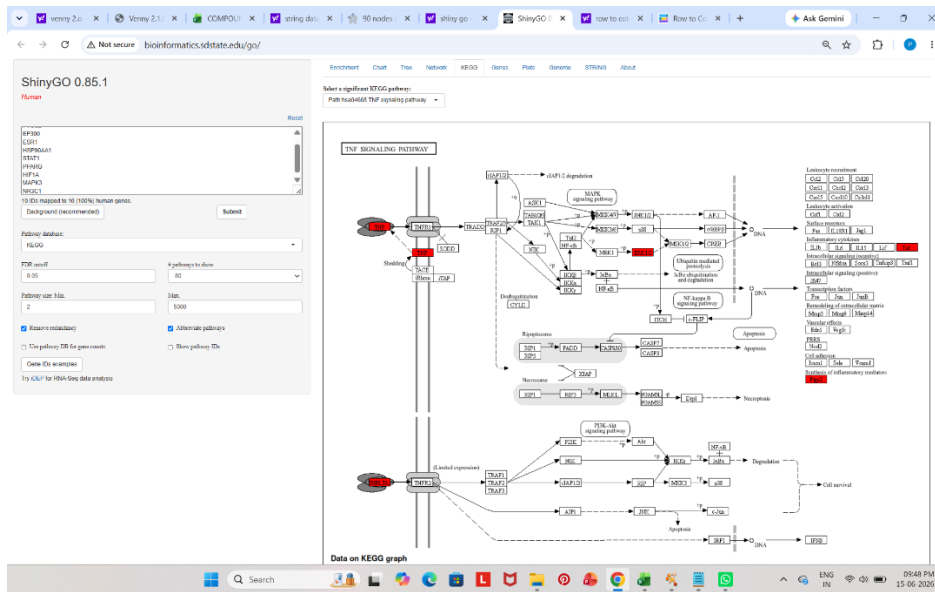


Figure 9: KEGG pathway enrichment analysis in shiny go web.

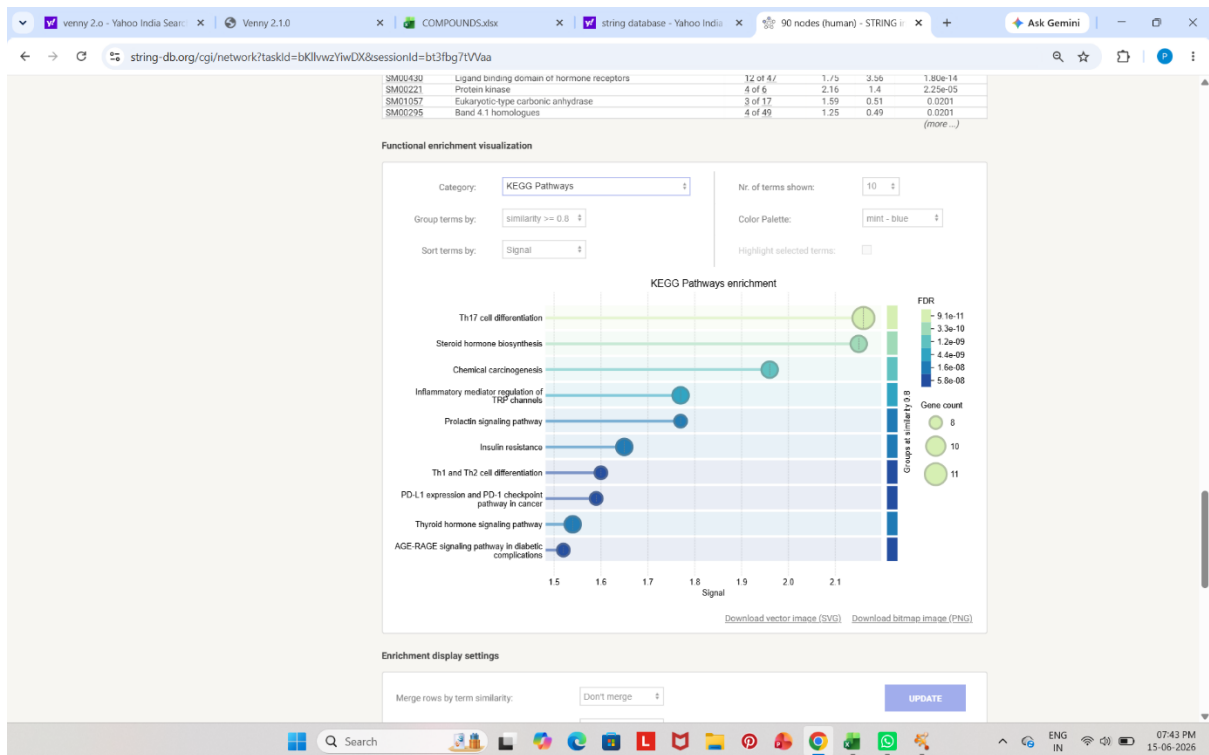


Figure 10: lollipop plotting for KEGG pathway:

Step 8: Protein Preparation

Protein preparation is a crucial step in molecular docking to ensure that the target protein is suitable for ligand binding analysis. The process involves cleaning the protein structure by removing unnecessary molecules, adding hydrogen atoms, assigning appropriate charges, and optimizing the structure to improve the accuracy of docking results.

PROCEDURE:

1. The three-dimensional structure of the target protein was downloaded from the Protein Data Bank (PDB).
2. The protein structure was opened in molecular visualization software.
3. Water molecules, co-crystallized ligands, and other unwanted heteroatoms were removed.
4. Polar hydrogen atoms were added, and Kollman charges were assigned to the protein.
5. The prepared protein structure was saved in PDBQT format for molecular docking.

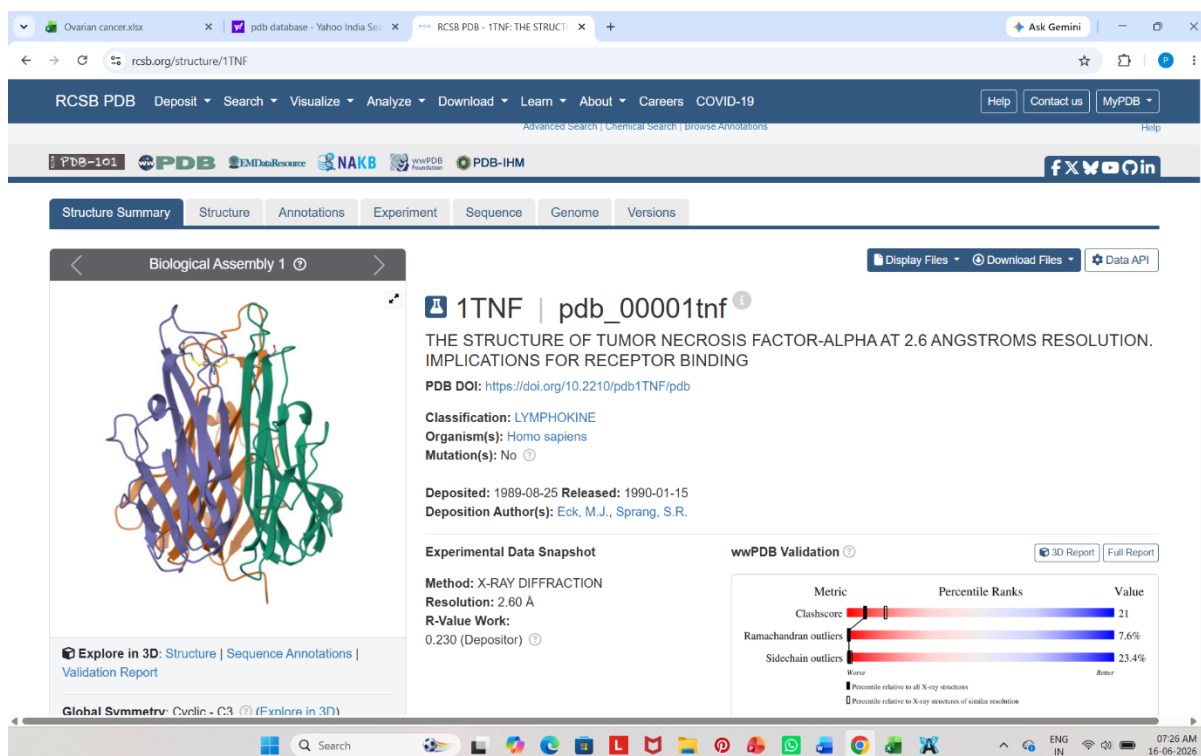


Figure 11: Protein structure from pdb

Step 9: Ligand Preparation in autodock

Ligand preparation is a crucial step in AutoDock to ensure that the ligand is in a suitable format for molecular docking. The process involves optimizing the ligand structure, adding hydrogen atoms, assigning Gasteiger charges, defining rotatable bonds, and converting the ligand into the PDBQT format required by AutoDock.

PROCEDURE:

1. The three-dimensional structure of nerol was obtained from the PubChem database in SDF format.
2. The ligand structure was converted to PDB format.
3. The PDB file was imported into AutoDock Tools (ADT).
4. Polar hydrogen atoms were added to the ligand.
5. Gasteiger charges were assigned, and non-polar hydrogens were merged.

6. Rotatable bonds were detected and defined automatically.

7. The prepared ligand was saved in PDBQT format for molecular docking.

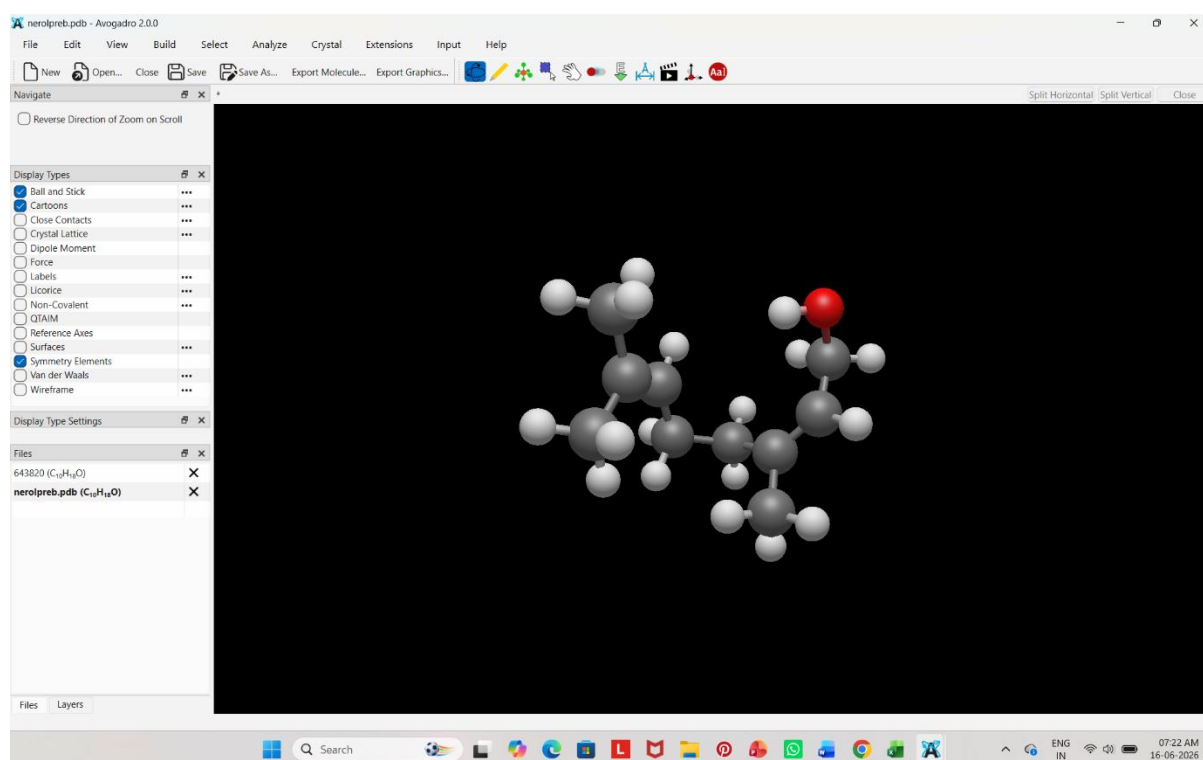


Figure 12: Ligand structure of nerol

Step 10: Molecular Docking:

Molecular docking is a computational technique used to predict the binding interaction between a ligand and a target protein. It estimates the binding affinity, binding orientation, and intermolecular interactions, thereby helping to evaluate the therapeutic potential of the ligand against the target protein.

PROCEDURE:

1. The prepared protein and ligand files in PDBQT format were imported into AutoDock
2. The docking grid box was defined to cover the active site of the target protein.
3. Docking parameters were set, and the docking simulation was performed using AutoDock

4. The binding poses were ranked based on their binding affinity (kcal/mol).
5. The best docking pose with the lowest binding energy was selected for further analysis.
6. The docked complex was visualized using molecular visualization software such as PyMOL or Discovery Studio Visualizer to analyze hydrogen bonds and other molecular interactions.

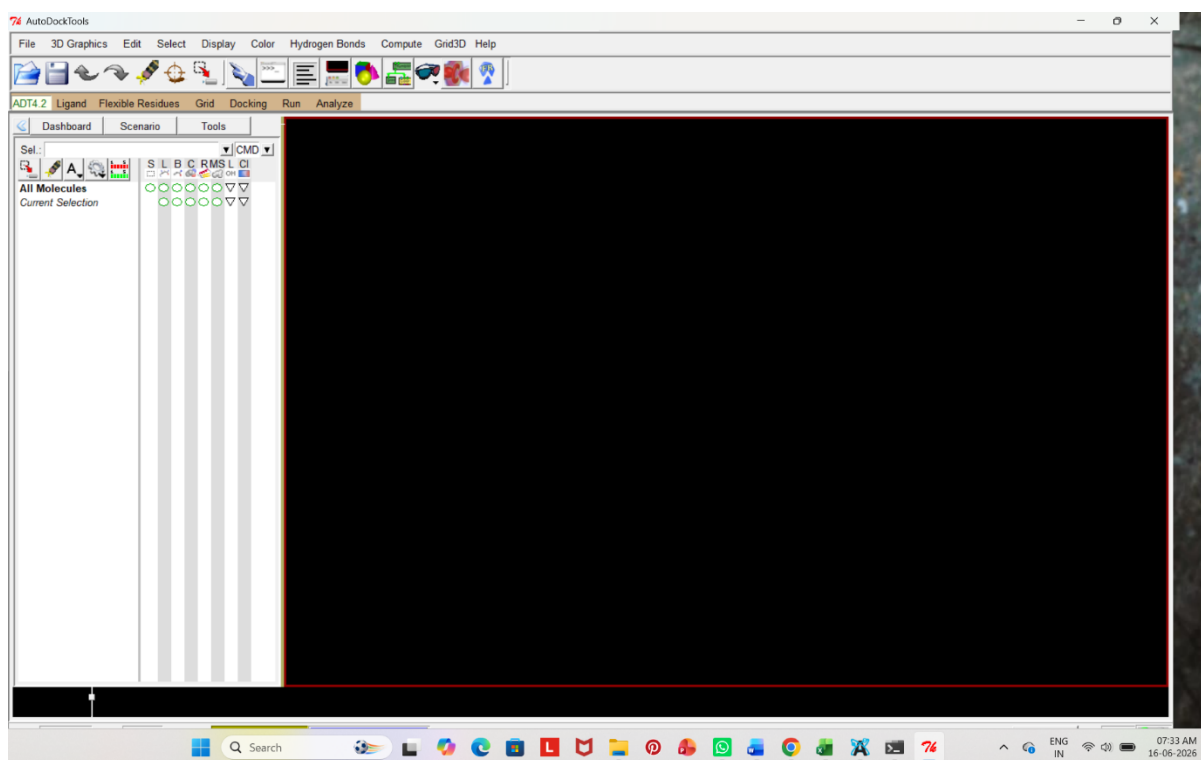


Figure 13: Docking setup window.

RESULTS OF DOCKING:

Molecular docking was successfully performed between nerol and the selected ovarian cancer target protein. The docking results provided the binding affinity and predicted binding pose of nerol within the protein's active site. The interactions observed suggest that nerol has the potential to bind effectively to the target protein, supporting its possible therapeutic role against ovarian cancer.

Step 11: Docking results visualization on Avogadro:

Avogadro is an open-source molecular visualization software used to examine the three-dimensional structure of docked protein–ligand complexes. It enables the visualization of ligand orientation, molecular geometry, and interactions within the protein binding site, facilitating the interpretation of molecular docking results.

PROCEDURE:

1. The best docking pose obtained from AutoDock was selected based on the lowest binding energy.
2. The docked complex was exported in a compatible file format (e.g., PDB).
3. The docked structure was opened in Avogadro.
4. The protein–ligand complex was visualized in three dimensions, and the binding orientation of nerol within the active site was examined.
5. Images of the docked complex were captured and saved for analysis and reporting.
6. 2D and 3D structures can be obtained.

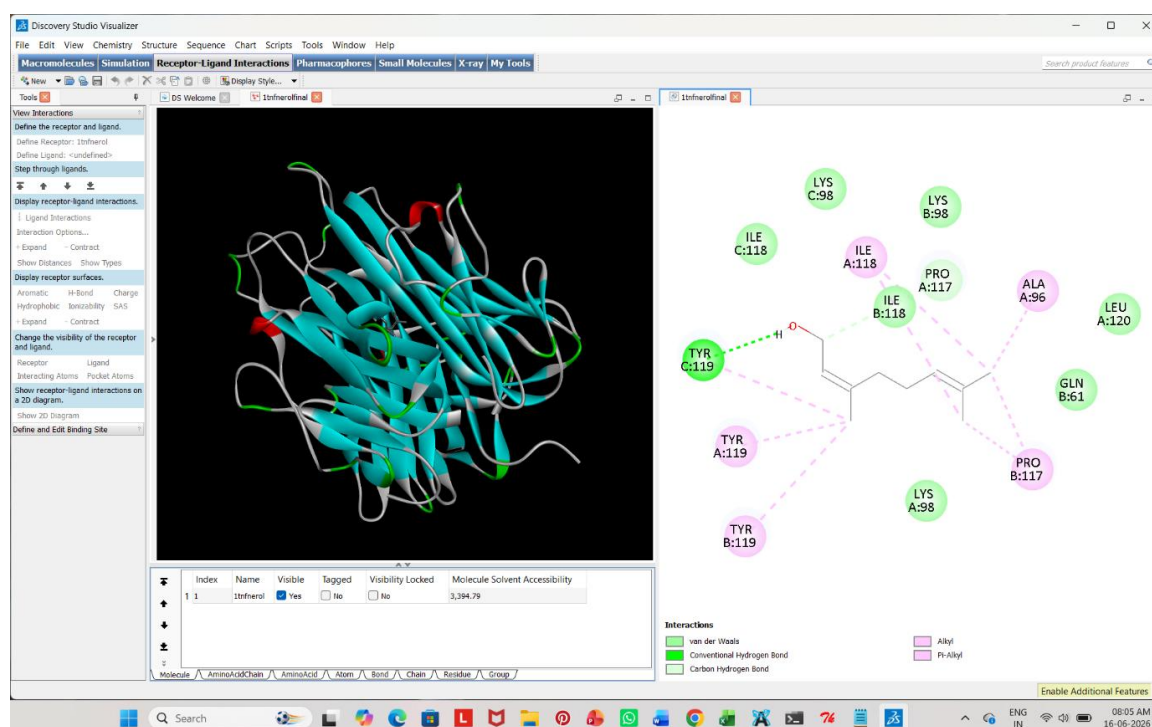


Figure 14: Docking result output.

7.SKILLS REQUIRED:

- Bioinformatics database searching
- Network pharmacology analysis
- Protein-protein interaction network construction
- Cytoscape visualization
- GO and KEGG enrichment analysis
- Molecular docking using AutoDock
- Protein-ligand interaction analysis
- Scientific report writing

8.RESULT AND DISCUSSION:

The phytochemical compounds of *Anethum graveolens* were identified, and nerol was selected as the lead compound based on the screening criteria. Potential molecular targets of nerol were predicted and compared with ovarian cancer-associated genes obtained from the GeneCards database. The overlapping targets indicated the possible therapeutic relevance of nerol in ovarian cancer.

A protein–protein interaction (PPI) network was constructed using the common target genes, and key hub proteins were identified through network analysis in Cytoscape. Gene Ontology (GO) and KEGG pathway enrichment analyses revealed that the common targets are involved in important biological processes and signaling pathways associated with cancer development, including cell proliferation, apoptosis, and signal transduction.

Molecular docking was performed between nerol and the selected hub protein. The docking results demonstrated favorable binding affinity and stable interactions between the ligand and the target protein. Visualization of the docked complex confirmed that nerol occupied the active binding site and formed interactions with key amino acid residues. These findings suggest that nerol may exert anti-ovarian cancer activity by modulating multiple molecular targets and signaling pathways, highlighting its potential as a promising natural therapeutic candidate.

9.CONCLUSION:

This study demonstrated the application of network pharmacology and molecular docking to investigate the therapeutic potential of nerol, a bioactive compound from *Anethum graveolens*, against ovarian cancer. The integrated analysis identified common target genes, key hub proteins, and significant biological pathways involved in the disease. Molecular docking further supported the ability of nerol to interact effectively with the selected target protein.

Overall, the findings indicate that nerol may possess anti-ovarian cancer activity through a multi-target and multi-pathway mechanism. Although the computational results are promising, further in vitro, in vivo, and clinical studies are required to validate its efficacy and safety before considering therapeutic applications.

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